



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI BHARATHI AGENCIES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

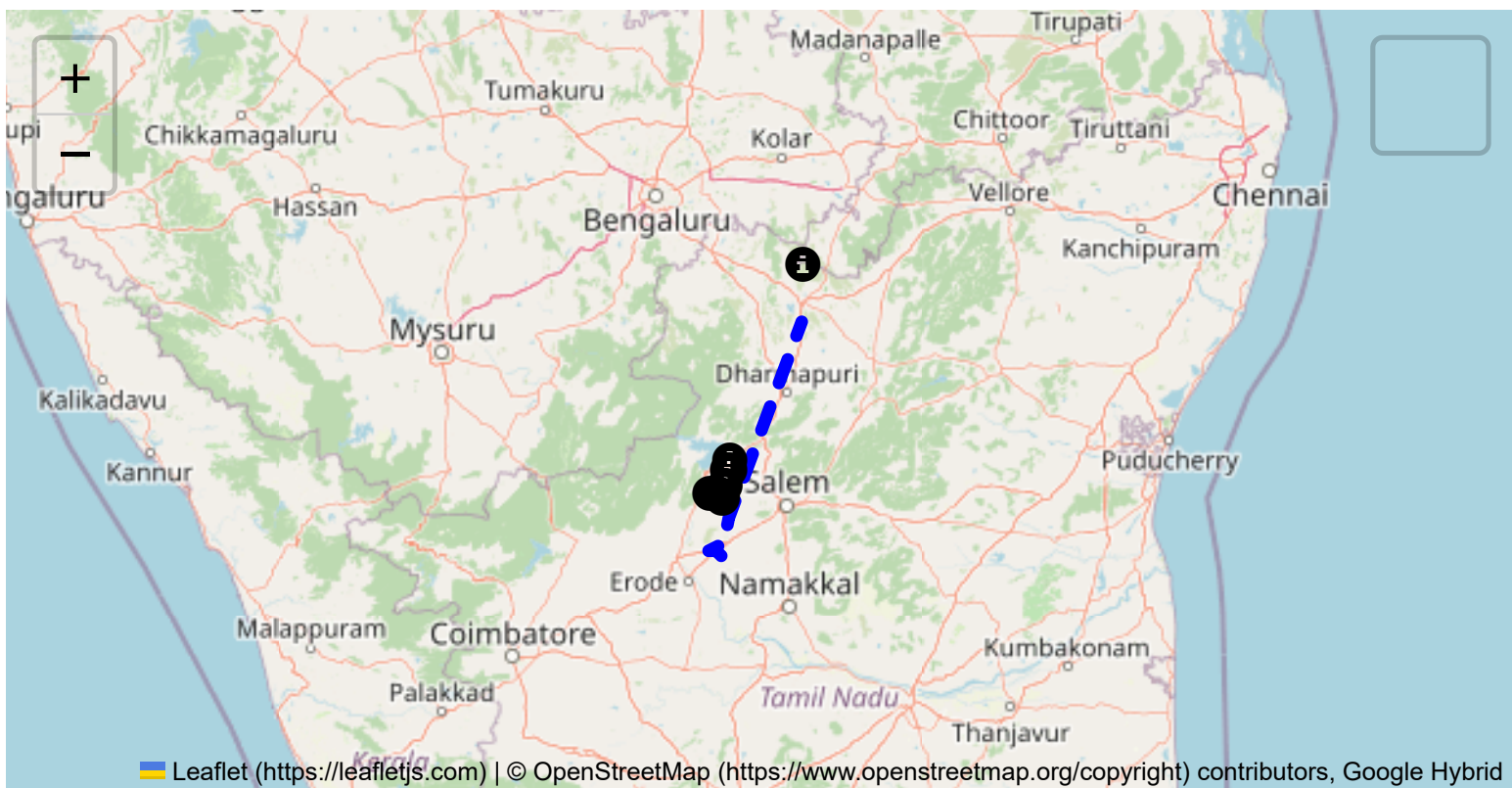
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

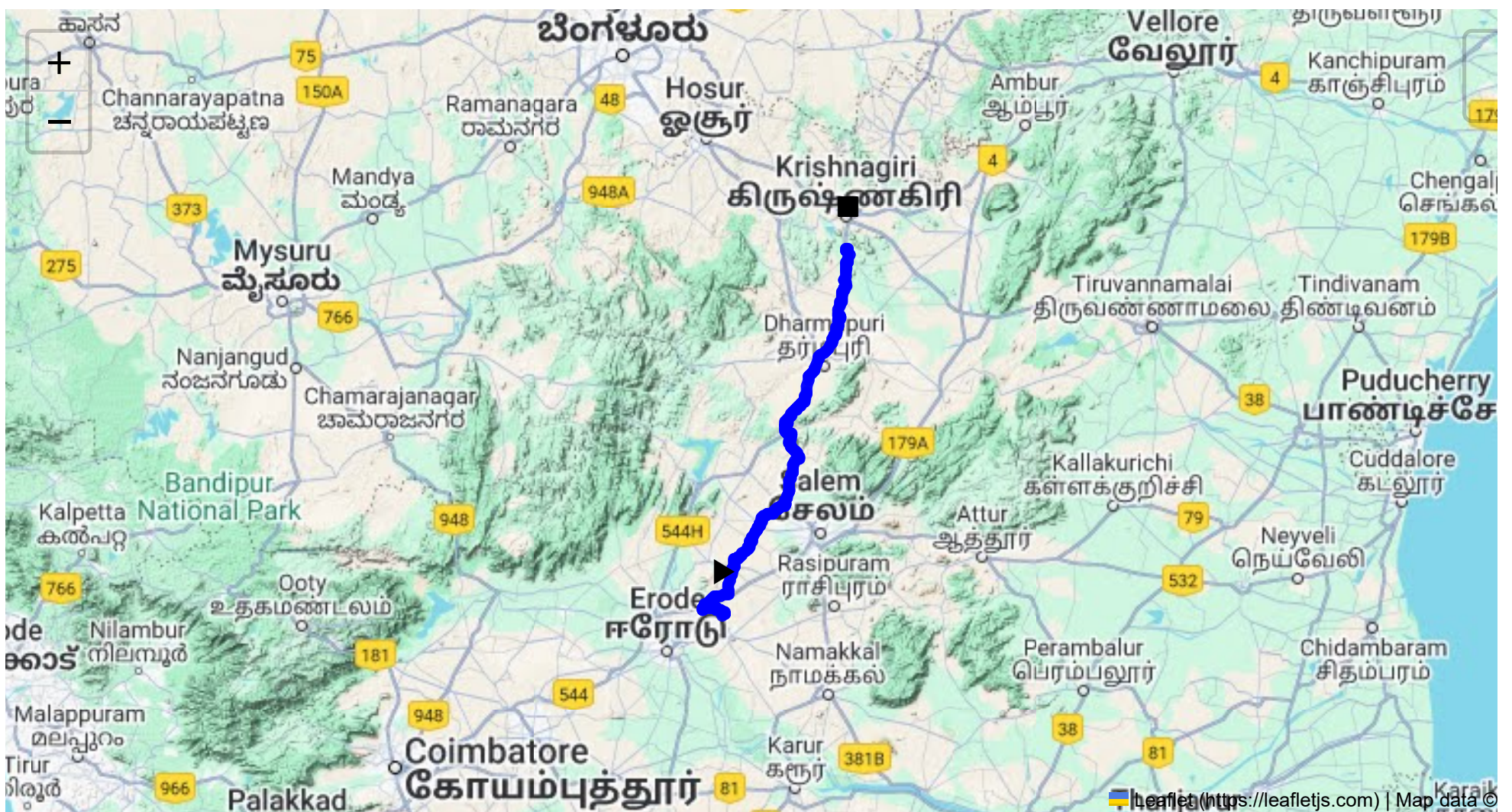
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 142.24 km
Estimated Duration: 2.8 hours
Adjusted Duration (Heavy Vehicle): 3.4 hours
Start: (11.4381, 77.8734)
End: (12.436803, 78.219443)



Welcome to the Journey Risk Management Study

1. **Overview of the Route Map:** The route from CVQF+23W, Sangagiri to 224, Salem Main Rd, Subramaniyapuram, covers approximately 142.24 kilometers through various towns and regions in Tamil Nadu, India. Key areas along the route include Sankari, Bhavani Main Road, and SH 86, passing through several small towns and rural areas. The path involves traveling on connecting roads and the SH 86 highway, providing access to local as well as longer routes.

2. **Typical Weather Conditions and Potential Weather-Related Hazards:** Tamil Nadu generally experiences three main seasons: summer (March to June), monsoon (July to October), and winter (November to February). Monsoon season presents the highest risk for weather-related hazards, such as flooding and reduced visibility due to heavy rains. Dry and hot summers could also lead to tire blowouts and heat stress for drivers.
3. **Analysis of Traffic Patterns:** Traffic congestion is likely to be heavier during morning (8 AM - 10 AM) and evening (5 PM - 8 PM) peak hours, particularly near towns and marketplaces. The city centers and main roads are most prone to congestion due to local business activities and transportation of goods. The vicinity of petrol bunks and skill crossings may also experience bottlenecks.
4. **Assessment of Road Quality and Infrastructure:** Road quality varies along the route, with regional roads typically narrower and potentially more worn than the SH 86 highway. Potholes and uneven terrain may be present, especially in less maintained segments near rural areas. Bridges and crossings are generally maintained but should be approached cautiously.
5. **Suggestions for Alternative Routes for Emergencies:** While the primary route is the most direct, traveling through NH 44 and connecting to SH or local roads generally provide alternative paths less influenced by local events. Keeping maps updated with real-time traffic apps can assist in emergency detours.
6. **Summary of Local Regulations Affecting Hazardous Material Transport:** Transportation of hazardous materials in Tamil Nadu requires adherence to local and national regulations, including obtaining necessary permits, adhering to speed limits, and ensuring proper labeling and containment of materials. Drivers must be trained and certified for handling hazardous goods.
7. **Overview of Historical Incidents:** Specific historical data on incidents involving heavy vehicles might be limited publicly, but general hazards have included road accidents due to over-speeding and poor road conditions, particularly during monsoon. Keeping in contact with local authorities for up-to-date information is critical.
8. **Environmental Considerations and Sensitive Areas:** The route passes through areas of agricultural significance and populated towns, where pollution could have serious impacts. Drivers carrying hazardous materials need to be aware of these areas to avoid spills and ensure substances are securely contained.
9. **Analysis of Communication Coverage:** While most of Tamil Nadu has good telecommunications coverage, there may be sporadic dead zones, particularly in rural or heavily forested regions. Drivers should ensure that communication devices are fully charged, and alternative contact options are available.
10. **Estimated Emergency Response Times:** Urban areas can expect faster emergency response times of 15-30 minutes due to proximity to services. In rural or remote areas, this could extend to 45-60 minutes. Awareness of local emergency service numbers and their stations' locations is essential.
11. **Overall Summary of Risk Assessment:** Overall, the route poses moderate risks due primarily to weather conditions, road quality variances, and traffic congestion during peak hours. Emergency preparedness, careful planning, and adherence to safety protocols are crucial to mitigating these risks

effectively. Enhanced driver training on local regulations and safe material handling can further ensure a safe journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43968, 77.87345	15 KM/Hr
1	Turn	High	11.44029, 77.87544	15 KM/Hr
2	Turn	High	11.44900, 77.87406	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	High	11.45571, 77.81702	15 KM/Hr
7	Turn	High	11.45615, 77.81677	15 KM/Hr
8	Turn	Medium	11.46916, 77.82431	30 KM/Hr
9	Blind Spot	Blind Spot	11.49395, 77.88537	10 KM/Hr
10	Turn	High	11.49465, 77.88488	15 KM/Hr
11	Turn	Medium	11.56096, 77.89931	30 KM/Hr
12	Turn	Medium	11.57649, 77.90610	30 KM/Hr
13	Turn	Medium	11.59214, 77.90656	30 KM/Hr
14	Turn	Medium	11.63833, 77.94818	30 KM/Hr
15	Turn	Medium	11.64171, 77.95184	30 KM/Hr
16	Turn	Medium	11.64243, 77.95198	30 KM/Hr
17	Turn	Medium	11.64721, 77.95118	30 KM/Hr
18	Turn	Medium	11.67870, 77.96846	30 KM/Hr
19	Turn	High	11.70227, 77.97864	15 KM/Hr
20	Turn	Medium	11.71715, 77.99969	30 KM/Hr
21	Turn	Medium	11.71921, 78.01133	30 KM/Hr
22	Turn	Medium	11.72737, 78.02248	30 KM/Hr
23	Turn	Medium	11.73775, 78.03465	30 KM/Hr
24	Turn	High	11.73651, 78.03878	15 KM/Hr
25	Turn	Medium	11.74189, 78.04660	30 KM/Hr
26	Turn	Medium	11.74309, 78.04725	30 KM/Hr
27	Turn	Medium	11.77961, 78.05775	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
28	Turn	Medium	11.86873, 78.08374	30 KM/Hr
29	Turn	Medium	11.87966, 78.08187	30 KM/Hr
30	Turn	Medium	11.92609, 78.06513	30 KM/Hr
31	Turn	Medium	11.93719, 78.05372	30 KM/Hr
32	Turn	High	11.95828, 78.04983	15 KM/Hr
33	Turn	Medium	11.95997, 78.04946	30 KM/Hr
34	Turn	Medium	11.96335, 78.05141	30 KM/Hr
35	Turn	High	11.96408, 78.05131	15 KM/Hr
36	Turn	Medium	11.96511, 78.04966	30 KM/Hr
37	Turn	High	11.96608, 78.04919	15 KM/Hr
38	Turn	Medium	11.96819, 78.05088	30 KM/Hr
39	Turn	Medium	12.17005, 78.16792	30 KM/Hr
40	Turn	Medium	12.19919, 78.18172	30 KM/Hr
41	Turn	Medium	12.22472, 78.19140	30 KM/Hr
42	Turn	Medium	12.24623, 78.19524	30 KM/Hr
43	Turn	Medium	12.25078, 78.19988	30 KM/Hr
44	Turn	Medium	12.25370, 78.20034	30 KM/Hr
45	Turn	Medium	12.29150, 78.19930	30 KM/Hr
46	Turn	Medium	12.40604, 78.22285	30 KM/Hr
47	Turn	Medium	12.42500, 78.22672	30 KM/Hr
48	Blind Spot	Blind Spot	12.44082, 78.21942	10 KM/Hr
49	Turn	Medium	12.43745, 78.21921	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
3	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
4	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
5	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
10	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
12	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
13	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
14	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
15	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
16	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
17	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
18	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
19	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
20	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
21	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
24	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
26	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
27	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
28	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
30	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
11	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
23	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
25	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
29	college	Laxminarayana Polytechnic College	12.0421277, 78.103203	30 km/h	Medium
31	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45764, 77.85219



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45571, 77.81702



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45615, 77.81677



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46916, 77.82431



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49395, 77.88537



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49465, 77.88488



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.56096, 77.89931



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57649, 77.90610



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.59214, 77.90656



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.63833, 77.94818



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64171, 77.95184



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64243, 77.95198



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64721, 77.95118



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67870, 77.96846



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70227, 77.97864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71715, 77.99969



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71921, 78.01133



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72737, 78.02248



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73775, 78.03465



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03878



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74189, 78.04660



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74309, 78.04725



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.77961, 78.05775



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86873, 78.08374



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.87966, 78.08187



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.92609, 78.06513



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.93719, 78.05372



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95828, 78.04983



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.95997, 78.04946



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96335, 78.05141



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96408, 78.05131



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.96511, 78.04966



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96608, 78.04919



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96819, 78.05088



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.17005, 78.16792



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.19919, 78.18172



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.22472, 78.19140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.24623, 78.19524



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25078, 78.19988



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.25370, 78.20034



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.29150, 78.19930



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.40604, 78.22285



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.42500, 78.22672



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 12.44082, 78.21942



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.43745, 78.21921
